

Question number	Answer	Additional guidance	Mark
7(a)	Choose an item: B is the correct - A is not the correct answer as only one statement is correct. - C is not the correct answer as only one statement is correct. - D is not the correct answer as only one statement is correct.		Computer (1)

Question number	Answer	Additional guidance	Mark
7(b)	A description that includes four of the following points: - reference to activation of (different) genes by {chemicals / hormones / transcription factor} (in specific areas of the nervous system) (1) - transcription of (three) genes to produce pre-mRNA (1) - post transcriptional (mRNA) modification (to produce 14 types of mRNA) (1) - causing translation (to produce receptors that bind serotonin) (1) - with different amino acid sequences (outside the receptor) (1)	accept ref to differential gene expression accept rearrangement of exons / alternative splicing / removal of some exons accept RNA splicing / mRNA isoforms produced accept different primary structure / polypeptide chain / tertiary structure	Expert (4)

Question number	Answer	Additional guidance	Mark
7(c)(i)	An answer that includes: - as the dose of MDMA increases the serotonin level increases (1) - calculation using data from graph (1)	ALLOW positive correlation accept as and with if they start with serotonin levels LOBF actual 0 – 5 mdma = 25 +/- 5 70 +/- 5 0 -10 mdma = 70 +/-5 65 +/- 5 0 -15 = mdma =105 +/-5 125 +/- 5 5 – 10 mdma = 45 +/-5 -5 +/- 5 5 – 15 mdma = 80 +/-5 55 +/- 5 10 – 15 mdma = 35 +/-5 60 +/- 5	Expert (2)

Question number	Answer	Additional guidance	Mark
7(c)(ii)	A calculation showing the following steps: - correct extrapolation from graph (1) - correct conversion to g cm^{-3} (1)	400 ± 4 $0.4 \text{ (g cm}^{-3}\text{)}$ accept $0.396 - 0.404 \text{ (g cm}^{-3}\text{)}$ units are not needed	Graduate (2)

Question number	Answer	Additional guidance	Mark
7(c)(iii)	An answer that includes three of the following points: • (digested) MDMA absorbed into blood (in mouth / gut/ intestinal tract) (1) • {carried in blood to brain /crosses blood brain barrier} (1) • (MDMA) binds to {neurones / receptors / proteins} • {stimulating the release of / increasing} serotonin (in CSF or brain) (1)		Expert (3)